

While I applaud Liu *et al.*¹ for their enterprising work, in the future, let's show readers the actual data alongside model estimates.

SUPPLEMENTAL INFORMATION

Supplemental information including two figures, experimental procedures, and further results and discussion can be found with this article online at <https://doi.org/10.1016/j.cub.2026.04.009>.

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DECLARATION OF INTERESTS

The author declares no competing interests.

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Letter

Response to Creel

Courtney B. Hilton^{1,*}, Jingxuan Liu², Samuel A. Mehr^{3,4}, and Erika Bergelson⁵

Our original paper¹ found effects of linguistic experience on music perception which generalized across a half-million speakers of fifty-four languages. In a critique, Creel² claims that our multilevel modelling approach biased the language-level estimates such that languages with small-sample sizes “spuriously appear to pattern consistently” compared to “raw” sample means. We respectfully disagree and counter this claim here.

First, Creel's “raw” mean approach needlessly restricts the data to participants without musical training, discarding 69% of the sample. This severely reduces statistical power for the small-N languages, where only 47% of observations remain; for Kinyarwanda and Ewe this leaves just four participants each — rendering sample means trivially unstable. While accounting for variation due to musical training is important, excluding a majority of the data is unnecessary: this can be handled statistically, as with our original analyses. Creel's approach also fails to adjust for imbalances in age, gender, and country. Thus, we believe these analysis decisions make the comparisons uninterpretable and were not justified.

Second, Creel also re-ran our multilevel model while excluding 98% of tonal language speakers, leaving 611 speakers of 13 tonal languages. She interpreted the resulting narrowly non-significant effect as evidence against robustness, arguing that 611 participants constitute a strong test because this is typically considered a large sample.

We find this misleading. A sample of 611 would be large for a non-hierarchical model assuming independent samples (which would find a tonal-language advantage: $\beta = 0.208$, $p < 0.001$). But this ignores how those 611 participants are sparsely sampled from 13 tonal languages and 53 countries. Our multilevel model accounts for both this sparseness and these interdependencies, providing a more realistic test of generalizability^{3,4}. Creel's decision to analyze only sparse data leaves this model underpowered: when

we conducted 100 simulations assuming the original effect size, only 68 detected the effect. Moreover, restricting the analysis to small-sample languages violates assumptions required for interpretable test statistics, including group exchangeability and unconditional sampling. Thus, Creel mistakes an underpowered test for a highly powered one and bases her conclusion on a p-value that is not statistically interpretable.

Third, Creel's permutation test contradicts her conclusion that our analysis was “circular”. It shows that the observed ranking of language estimates is unlikely under the null. However, this understates how unlikely the result is because rankings are the least sensitive metric available. When we repeated this test, we found that only 3 of 1,000 simulations produced a mean effect for small-N languages as large or larger than observed in the real data ($p = 0.003$; Figure 1).

We note one nuance, however. As shown in Figure 1, the permuted estimates are not centered on zero but shifted rightward. As Creel correctly notes, this reflects shrinkage — a core feature of multilevel models — whereby partially pooled estimates are pulled toward the mean. This is a feature, not a bug. Accepting a small amount of bias to reduce estimator variance is a standard part of modern statistical estimation, and counterintuitively, generally produces more, not less, interpretable estimates⁵. This is especially so relative to “raw” means, which face two problems: (1) high estimator variance, making them unreliable for small-n languages; and (2) a tendency to overfit the data. A partially pooled estimator addresses both issues.

To demonstrate this here, we simulated individual and language-level variation matched to the distributional properties, demographics, and sample sizes of the real data and attempted to recover simulated language scores using either sample means or partially pooled estimates, repeating this 100 times. The results were clear (Figure S1 in Supplemental information). Although less biased, a sample mean estimator was too variable to be useful, producing estimates ranging over an interval approximately 20 times larger than the original effect. By contrast, the partially pooled estimator was an



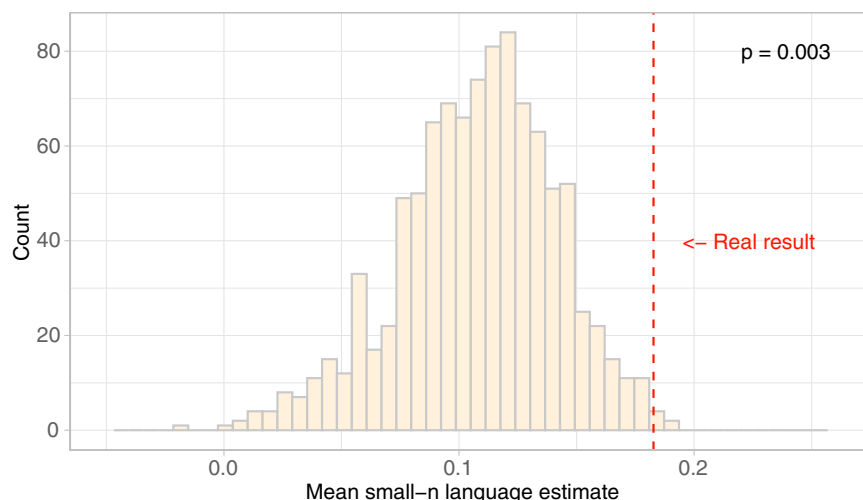


Figure 1. Original tonal-language advantage in small-sample languages is unlikely to be produced by chance.

The histogram shows the results of a permutation test evaluating Creel's claim that shrinkage alone determines the estimated tonal-language advantage in the 13 smallest-N languages. The x-axis shows the average estimated tonal-language advantage for these languages under the null; the y-axis shows the number of simulations in each histogram bin. The red dashed vertical line marks the estimate obtained from the real data. Across 1,000 permutation replications, only 3 yielded values as large as or larger than the observed estimate ($p = 0.003$), indicating that the observed result is highly unlikely to arise from the model under the null.

order of magnitude more accurate across simulations (MSE for sample mean = 0.109; MSE for partially pooled = 0.015, $p < 0.001$, two-sample t-test) and the small amount of bias approached zero across languages.

Although it is unlikely that African tonal languages fully lack a tonal-language advantage, we agree with Creel that the effect in such languages could be smaller. Our original paper did not preclude this possibility, contra Creel's claim that our model encoded a "tone language consistency hypothesis". Neither our results nor our modelling logic implied such a "unitary tone-language group". Thus, we stand by our conclusion¹ that "across a range of geographic and demographic contexts, linguistic experience alters music perception ability in reliable (but not unitary) ways".

Critically, if African tonal languages do have systematically smaller effect sizes, partial pooling would only modestly overestimate them. To demonstrate this, we simulated scenarios in which small-sample tonal languages differed from other tonal languages by 0.05, 0.10, or 0.20 z-score units. In every case, partially pooled estimates were more accurate for all languages (all p s < 0.001 ; Figure S2). Even in the most extreme

scenario where African tonal languages showed no average advantage over non-tonal languages, partial pooling was more accurate and gave estimates more similar to the simulated truth than the tonal average they were pulled towards.

We return to what we see as the core disagreement. We agree that distinguishing data from models is essential, and we should have communicated our modelling assumptions and descriptive summaries more clearly. However, as we have shown, aspects of Creel's re-analyses undermine her critique. Her argument also places undue weight on whether model estimates reproduce the patterns in raw data. This, to us, is misguided. Raw data rarely reveal relationships of interest⁶, particularly in large, heterogeneous datasets. In such contexts, principled multilevel modelling is essential for robust, generalizable inference^{3,7}, and model-based estimates should not be expected to mirror descriptive means.

In conclusion, although we disagree with Creel's critique, we agree further work is needed and that more transparent modeling details benefit everyone. We share her intuition that effects of language experience on music processing may differ for African

tonal languages, compared to East-Asian tonal languages, and welcome future research exploring this question and/or refining the methodological issues raised here.

SUPPLEMENTAL INFORMATION

Supplemental information including two figures can be found with this article online at <https://doi.org/10.1016/j.cub.2026.04.007>.

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We thank the reviewers for thoughtful and detailed engagement, especially Bodo Winter who suggested additional analyses we were unable to include due to space constraints. These extra figures, including combined raw and model-based estimate plots, and language-family-adjusted results can be found at https://github.com/themusiclab/creel_response.

AUTHOR CONTRIBUTIONS

Conceptualization, C.B.H., J.L., S.A.M., and E.B.; writing — original draft, C.B.H.; writing — review & editing, C.B.H., J.L., S.A.M., and E.B.; formal analysis, visualization, and software, C.B.H.; supervision, E.B. and S.A.M.

DECLARATION OF INTERESTS

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